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Editorial Board
PLoS Computational Biology

Dear Editors,

We are pleased to submit our manuscript, “**Hyperdimensional Active Inference: Free Energy Principle in Vector Symbolic Architectures**,” for consideration as a Research Article in PLoS Computational Biology.

Summary. This paper presents Hyperdimensional Active Inference (HAI), the first integration of the Free Energy Principle with Hyperdimensional Computing (HDC). By reformulating variational free energy in 16,384-dimensional hypervector space, we achieve $O(d)$ complexity where existing active inference implementations require $O(n^2)$ – $O(n^3)$ matrix operations. The work introduces precision-weighted binding—a novel HDC operation for confidence-modulated feature composition—and derives eight interpretable motor commands directly from expected free energy minimization.

Key contributions:

1. **Computational efficiency:** $7.9\times$ total speedup over pymdp ($1.9\times$ belief inference, $15.8\times$ action selection) on standard benchmarks (T-Maze, Grid World), with success rates improving from 10–16% to 88–100%.
2. **Cross-domain validation:** 17 benchmarks spanning neuroscience (*C. elegans* connectome analysis, EEG seizure detection), ethics (92.9% on ETHICS benchmark via compositional moral algebra—no pretrained language models), signal processing (94.5% speaker identification), and federated learning (Byzantine tolerance validated to 34%).
3. **Mathematical foundations:** 14 rigorous mathematics modules grounding HDC in established theory (information theory, IIT 3.0, Riemannian geometry, tensor algebra, causal calculus, and more).
4. **Open-source implementation:** Complete Rust codebase ($\sim 338\text{K}$ LOC, 3,200+ tests), all benchmarks reproducible via standalone examples.

Relevance to PLoS Computational Biology. This work addresses a core challenge in computational neuroscience: making the Free Energy Principle computationally tractable at scale. The biological validation (*C. elegans* connectome, EEG analysis) and the brain-inspired HDC substrate align directly with the journal’s scope. The open-source codebase and reproducible benchmarks align with PLoS’s commitment to open science.

Novelty. While Liquid AI (LFM2/LFM2.5) validates liquid neural networks commercially and FedHDC demonstrates federated HDC, no existing system integrates active inference with hyperdimensional computing. Our HDC moral algebra achieves 92.9% on ETHICS using only vector operations—a fundamentally different approach from large language models.

Reproducibility. All benchmarks are reproducible via standalone Rust examples with deterministic seeds. The complete codebase, raw timing data, and supplementary materials are available at <https://github.com/Luminous-Dynamics/symthaea>.

Data Availability. All synthetic benchmarks are generated from code. External datasets used (Sleep-EDF, LibriSpeech, ETHICS, Social Chemistry 101) are publicly available under open licenses.

Sincerely,

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